IU Mobile App: Dedicated Campus Transit & Navigation Hub

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Product Target - IU Mobile iOS & Android Applications

Target Launch Window - December 2026

1. Overview

This spec proposes a dedicated transit and navigation hub inside the IU Mobile app for riders on the IU Campus Bus and Bloomington Transit. It replaces the current unreliable bus tracker with traffic-adjusted ETAs, an accurate count of active buses, building-to-stop search, labeled stops and a decluttered map. The goal is to keep students and faculty in the IU Mobile app instead of sending them to Google Maps, ETA Spot or rideshares. Target launch is December 2026.

2. Problem Statement

Students and faculty at Indiana University Bloomington rely heavily on campus and city transit to commute across a large, spread-out campus. However, the current transit feature inside the IU Mobile app provides an unreliable and frustrating user experience, forcing riders to abandon the native app for external alternatives.

Core User Pain Points

- **Inaccurate & Lagging ETAs:** Bus arrival predictions freeze or stall (e.g., displaying "5 minutes away" for over 15 minutes) due to delayed GPS sync and a lack of live traffic accounting.
- **"Ghost Buses" & Unclear Fleet Capacity:** Buses frequently vanish from maps or fail to arrive, and users cannot see how many total active buses are currently servicing a given route.
- **Missing Campus Spatial Context:** Stops appear as plain, unlabelled black dots on the map. Users cannot easily connect bus routes to actual campus buildings (e.g., Hodge Hall, Wells Library), forcing them to open Google Maps to cross-reference location details.
- **Cluttered Map Interface & Deep Menus:** Multiple bus icons stack directly on top of each other, obscuring active routes. Filtering requires navigating multi-step drawer menus instead of quick-access toggles.

Business & Product Impact: Due to these systematic failures, students resort to opening ETA Spot in web browsers, switching to third-party navigation apps, or calling expensive rideshares (Uber/Lyft) to avoid being late for classes. This results in high user churn and low trust in the core IU Mobile application ecosystem.

3. Goals & Success Metrics

Draft added to fill the missing section. Only the two targets taken from your spec are firm; the rest need a baseline from IU transit data.

Goal	Metric	Target
Trustworthy arrival times	Gap between predicted and actual bus arrival	To be set after measuring the current baseline
No ghost buses	Buses shown on the map with no GPS ping for >3 minutes	0 (from the data invalidation rule)
Smooth live map	Bus location refresh interval	5–10 seconds (from the performance spec)
Fewer trips to other apps	Transit sessions that end in Google Maps, ETA Spot or a rideshare	To be set after measuring the current baseline
Faster route finding	Time from opening the app to choosing a stop for a building	To be set after user testing

4. Target Audience & Personas

Primary Persona: The Daily Commuter

- **Demographics:** Undergraduate/Graduate student living off-campus or in distant dorms (e.g., McNut, Briscoe).
- **Core Goal:** Get from their starting location to back-to-back classes (e.g., Hodge Hall, Kelley School of Business) as quickly as possible without missing a bus.
- **Current Friction:** Frustrated by "ghost buses" showing 5-minute arrival times for 20 minutes, causing them to be late for exams or lectures.
- **Key Need:** Hyper-accurate ETAs, traffic delay adjustments, and active bus counts so they know whether to wait or walk.

Secondary Persona: The Casual Rider

- **Demographics:** Freshman, campus visitor, or student who usually walks/scooters but relies on buses during bad weather or long cross-campus trips.
- **Core Goal:** Quickly identify which bus route takes them directly to a specific building without having to memorize complex route letters.
- **Current Friction:** Confused by unlabelled black dots on the map and forced to switch to Google Maps to find out which stop is closest to their target building.
- **Key Need:** Direct building-to-stop search, persistent stop labels on the map layer, and simple route directions.

5. Functional Requirements

Feature ID	Feature Name	Description	Priority
FR-01	Building-to-Stop Search	It allows searching by campus building name (e.g., "Wells Library"). Auto-selects the closest bus stop and recommends optimal routes.	P0
FR-02	Traffic-Adjusted ETAs	Recalculate arrival predictions using live GTFS-RT position feeds combined with real-time Bloomington traffic congestion data.	P0
FR-03	Accurate Fleet Counter	Display total count of currently operating, active buses on each route card	P0
FR-04	Ghost Bus Purge	Automatically flag bus map markers if their GPS signal drops or freezes for >3 minutes	P1
FR-05	Persistent Map Labels	Render legible text labels on major bus stops across the map without requiring users to tap black dots	P1
FR-06	Bus Stacking Clustering	Group overlapping bus icons on identical route segments into a single dynamic cluster badge	P1
FR-07	One-Tap Route Chips	Horizontal, scrollable filter chips at the top of the screen (All, Route F, Route E, BT City) for instant map filtering.	P1

6. Technical Architecture & Data Scope

1. Data Sources & APIs

- **GTFS-Realtime (GTFS-RT) Feeds:** Primary data ingestion for live vehicle positions, route shapes, and stop sequences covering both IU Campus Bus and Bloomington Transit lines.
- **GeoJSON Campus Overlay:** Static geographic spatial dataset containing boundary outlines and entrance coordinate points for IU Bloomington buildings (e.g., Hodge Hall, Wells Library, IMU).
- **Traffic & Congestion Layer:** Integration with live traffic APIs (e.g., Google Maps Platform / Mapbox Traffic) to calculate dynamic delay vectors on key campus arteries (10th St, Fee Ln, 3rd St).

2. System Performance & Edge Handling

- **Location Ping Frequency:** Bus GPS locations update every 5–10 seconds to eliminate icon lagging across map tiles.
- **Data Invalidation Rules:** If a vehicle endpoint stops pinging for >180 seconds (3 minutes), the system marks the bus as "Offline" and removes it from the visible map interface to prevent "ghost bus" confusion.
- **Offline Caching & Fallback:** Local caching of static route paths and stop schedules ensures the app remains responsive during poor cellular reception in dense campus basements or lecture halls.

7. UX/UI & Screen Layout Specifications

Screen Structure & Visual Layout

- **Top Navigation Bar (Sticky)**
  - **Search Input Field:** Full-width search bar displaying placeholder text *"Search campus building or stop (e.g., Hodge Hall, Wells Library)"*.
  - **Filter Chips Row:** A horizontally scrollable row of single-tap pill buttons located directly below the search bar. Options include All, Route F, Route E, Route B, and Bloomington Transit. Selecting a pill filters the map to display only that specific route.
- **Main Interactive Map View**
  - **Building Overlay Layer:** Displays light grey 2D/3D outlines of major IU campus buildings with legible name labels.

    - **Bus Stop Markers:** Replaces plain black dots with distinct stop pins. Major high-traffic stops feature persistent, visible text labels (e.g., *"10th & Fee"*) at standard zoom levels.
    - **Vehicle Location Badges:** Live bus icons rendered in high-contrast route colors (e.g., Purple for Route F).
    - **Vehicle Cluster Badges:** When two or more buses overlap on a route segment, they merge into a single circular badge displaying the quantity (e.g., "3x F") to prevent visual clutter. Tapping expands the cluster.
- **Bottom Draggable Sheet (Card Overlay)**
    - **Default / Collapsed View:** Displays a "Nearest Stop" card based on the user's live GPS, showing the stop name, walking time, and live ETAs for incoming buses.
    - **Expanded View (Swiped Up):** Displays a list of all active routes. Each route card shows:
        - Route Letter & Name
        - Total Active Buses currently running
        - Chronological list of upcoming stops with nearby landmark tags (e.g., *"Stop: Fee Lane → Serves Kelley School"*)

Core User Interactions & State Behaviors

- **Searching for a Building:** Typing "Hodge Hall" automatically drops a target pin on Hodge Hall, highlights the nearest bus stop, and draws a dashed walking path from the stop to the building entrance.
- **Tapping a Bus Icon:** Opens a pop-up mini card showing bus ID, speed/status, live capacity (if available), and real-time traffic delay status (e.g., *"+2 mins due to 10th St traffic"*).
- **Handling Stale Data:** If a bus fails to send a GPS ping for >3 minutes, its icon fades out and disappears from the map view automatically to eliminate "ghost bus" confusion.